

Clémence SEBE

Third-year doctoral student at LISN,
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ACADEMIC BACKGROUND

PhD Student

2023 - PRESENT

LISN, UNIVERSITÉ PARIS-SACLAY

Thesis title: *Information Extraction from Scientific Literature: A Case Study on Bioinformatics Workflows*

Supervised by Sarah Cohen-Boulakia, Olivier Ferret, and Aurélie Névéol

Doctoral Stay

2025

HUMBOLDT-UNIVERSITÄT, BERLIN

Three-month internship within the Knowledge Management in Bioinformatics team

Under the supervision of Ulf Leser

Teaching Assignment (PhD)

2023 - 2026

UNIVERSITÉ PARIS-SACLAY (3 years, 3*64h)

Advanced Functional Programming, Introduction to Data Science

Research Intern

2023

LISN, UNIVERSITÉ PARIS-SACLAY (5 months)

Concordance between the description of bioinformatics workflows in scientific publications and their implementation in the associated code

Research Intern

2022

LISN, UNIVERSITÉ PARIS-SACLAY (3 months)

Extracting Information on Scientific Workflows from Literature

Study and Research Project

2021-2022

LISN, UNIVERSITÉ PARIS-SACLAY (9 months, 2 days per week)

Analysis of Processes in Nextflow Workflows and

Modeling and Implementation of a Database Compiling All Information on Workflows

UNIVERSITY DEGREES

Master's Degree in Computational Biology: Analysis, Modeling, and Engineering of 2021 - 2023

Biological and Medical Information

UNIVERSITÉ PARIS-SACLAY

Double Bachelor's Degree in Computer Science and Mathematics

2018 - 2021

UNIVERSITÉ PARIS-SACLAY

TEACHING

Volumes are indicated in equivalent tutorial hours (HETD).

Title	Audience	Year	Volume
Advanced Functional Programming	L3	2023-2026	Tutorials: 24
Introduction to Data Science	L1 & CPES1	2023-2026	Tutorials: 2*24
Introduction to Computer Science	L1	2023-2024	Tutorials: 10

PUBLICATIONS

International Peer-Reviewed Conferences

- **Sebe, Clémence**, Sarah Cohen-Boulakia, Olivier Ferret, and Aurélie Névéol. Extracting Information in a Low-Resource Setting: Case Study on Bioinformatics Workflows. In *Advances in Intelligent Data Analysis XXIII*, pages 274–287, Cham, 2025. Springer Nature Switzerland

National Peer-Reviewed Conferences

- Marie Schmit, Melvin Selim Atay, Khalid Belhajjame, Ulysse Le Clanche, Emmanuel Coquery, Olivier Dameron, Fabien Duchateau, Alban Gaignard, Mouna El Garb, Jaffar Gura, Nicolas Lumineau, George Marchment, Camille Maumet, **Sebe, Clémence**, Frédéric Lemoine, Sarah Cohen-Boulakia, and Hervé Ménager. ShareFAIR-KG, a centralised knowledge base of scientific workflows. In *JOBIM 2026 - Journées Ouvertes en Biologie, Informatique et Mathématiques*, Strasbourg, France, juin 2026
- **Sebe, Clémence**, Sarah Cohen-Boulakia, Olivier Ferret, and Aurélie Névéol. Extraction d’entités nommées décrivant des chaînes de traitement bioinformatiques dans des articles scientifiques en anglais. In *35èmes Journées d’Études sur la Parole (JEP 2024) 31ème Conférence sur le Traitement Automatique des Langues Naturelles (TALN 2024) 26ème Rencontre des Étudiants Chercheurs en Informatique pour le Traitement Automatique des Langues (RECITAL 2024)*, volume 1 : articles longs et prises de position, pages 422–434, Toulouse, France, juillet 2024. ATALA & AFPC
- **Sebe, Clémence**, Aurélie Névéol, Sarah Cohen-Boulakia, and Alban Gaignard. Extraction d’informations sur les workflows scientifiques à partir de la littérature. volume Extraction et Gestion des Connaissances, RNTI-E-39, page 313, janvier 2023

International Journals

- Marine Djaffardjy, George Marchment, **Sebe, Clémence**, Raphaël Blanchet, Khalid Belhajjame, Alban Gaignard, Frédéric Lemoine, and Sarah Cohen-Boulakia. Developing and reusing bioinformatics data analysis pipelines using scientific workflow systems. *Computational and Structural Biotechnology Journal*, 21:2075–2085, janvier 2023

Book Chapter

- Sarah Cohen-Boulakia, Frédéric Lemoine, George Marchment, Marine Djaffardjy, Alban Gaignard, **Sebe, Clémence**, and Khalid Belhajjame. Reuse and reproduce bioinformatic pipelines using scientific workflow systems. In Berlin Universities Publishing, editor, *Workflow systems for large-scale scientific data analysis*. Berlin Universities Publishing, 2026

Submitted, Under Review

- **Sebe, Clémence**, Olivier Ferret, Aurélie Névéol, Mahdi Esmailoghli, Ulf Leser, and Sarah Cohen-Boulakia. Supporting Workflow Reproducibility by Linking Bioinformatics Tools across Papers and Executable Code, mars 2026. arXiv:2603.08195 [cs]

OTHER RESEARCH-RELATED ACTIVITIES

PhD Student Representative at LISN <i>Participation in the laboratory council, organization of events for PhD students (post-thesis seminar, new lab members' welcome days, seminar on how to prepare a CSI)</i>	2024-2025
Invited Oral Presentation at the Scientific Animation Day organized by the Bioinformatics Axis of Biogenouest <i>Extraction d'informations sur les workflows scientifiques à partir de la littérature anglaise: premiers résultats, May 24, 2024, Rennes</i>	2024
Presentations at ShareFAIR Project Study Days <i>Extracting and Linking Workflow Information between Bioinformatics Literature and Source Code, December 10-11, 2025, Institut Pasteur</i> <i>BioToFlow corpus, November 28-29, 2024, Institut Pasteur</i> <i>Extraction de noms d'outils bioinformatiques dans la littérature scientifique, November 20-21, 2023, Université Paris-Saclay, LUMEN</i>	2023 - 2026
Presentations of PhD Work during Team Seminars at LISN <i>Bioinformatics Team, January 12, 2023, and February 5, 2026</i> <i>Reading Group of the Language Sciences and Technologies Department, September 27, 2024</i>	2023-2026
Contributed to the organisation of the Third Edition of the IA/Data Science Paris-Saclay Students Day by DataIA	2023
Presentations at the Junior Multidisciplinary Congress of Université Paris-Saclay <i>Extraction d'Informations sur les Workflows Scientifiques à partir d'Articles en Anglais, June 8, 2023, ENS Paris-Saclay</i> <i>Réutilisation de Workflows d'Analyse Bioinformatique, June 2, 2022, ENS Paris-Saclay</i>	2022-2023
Mentoring for M1 Students <i>Tutoring for Master 1 students in Bioinformatics and Biostatistics, 1 hour per week, supported by the DATA-IA Institute, Université Paris-Saclay, Orsay</i>	2022-2023
Presentation at R2P2 (Analyse de données multi-modales pour les Pathologies complexes par la conception et l'implémentation de Protocoles Reproductibles et Réutilisables) Project Study Days	2021

OUTREACH ACTIVITIES

Film-Debate <i>Participation in a Film-Debate with high school students (11th and 12th grade) from Lycée Blaise Pascal, Orsay on April 2. After the screening of the film "Le Théorème de Marguerite", a debate on the place of women in science</i>	2026
Speed-Meeting <i>Ambition Sciences pour elles, mentoring program with high school girls from Lycée Marceau in Chartres, March 12</i> <i>Career guidance for gender equality, with 9th-grade students from Collège Charles Péguy, Palaiseau, December 15</i>	2026

Creation of a Board Game for the Science Festival

2025

Operation S.A.N.T.É. – a board game to discover AI in healthcare

CO-SUPERVISION OF INTERNS

- 2026 Sukhjot Kaur (M1 - Université Paris-Saclay) : **Development of a Web Interface for the CoPaLink Tool**, with Sarah Cohen-Boulakia, Olivier Ferret, Aurélie Névéol.
- 2025 Anjali Badlani (M2 - University of Potsdam) : **NLP-based Workflow Analysis in Snake-make Repositories**, with Sarah Cohen-Boulakia.

DEVELOPED TOOLS AND RESOURCES

CoPaLink

CoPaLink is a tool designed to automatically extract the names of bioinformatics tools mentioned in a scientific article describing a Nextflow workflow, as well as in the executable source code of the workflow. A linking step between the bioinformatics tool names from both sources is then performed.

GitLab repository: <https://gitlab.liris.cnrs.fr/sharefair/copalink>

COMPUTER SKILLS

Programming Languages Mastered	:	PYTHON, SQL, BASH, L ^A T _E X.
Other Known Languages	:	C, JAVA, OCAML.
Databases	:	PostgreSQL.
Other Tools	:	R, Github.